

EcoShop Capstone - Deployment Guide

Overview

EcoShop is a static website with no backend requirements, making it trivial to deploy to any static hosting platform. This guide covers three popular options: Vercel, Netlify, and Cloudflare Pages.

Pre-Deployment Checklist

Code Quality

- ☐ No console errors or warnings
- ☐ All features tested locally
- ☐ Responsive design verified (320px, 768px, 1024px+)
- ☐ Cross-browser compatibility confirmed
- ☐ No external API dependencies

Repository Setup

- ☐ Project pushed to GitHub

- ☐ README.md configured
- ☐ .gitignore created
- ☐ All necessary files included:
 - capstone-index.html
 - capstone-app.js
 - storage.js
 - store.js
 - Documentation files

Performance

- ☐ Lighthouse score ≥ 90
 - ☐ PageSpeed score ≥ 90
 - ☐ Bundle size $< 100\text{KB}$
 - ☐ Load time $< 2\text{s}$
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Option 1: Vercel (Recommended)

Why Vercel?

- Zero-config deployment
- Automatic deployments on push
- Generous free tier
- Excellent performance globally
- GitHub integration included

Step-by-Step Deployment

1. Prepare Repository

```
cd ecoshop-capstone

# Create .gitignore
echo "node_modules/" >> .gitignore
echo ".DS_Store" >> .gitignore
echo "*.log" >> .gitignore

# Initialize git (if needed)
git init

# Add all files
git add .

# Commit
git commit -m "Initial commit: Capstone project"

# Add remote and push
git remote add origin
https://github.com/yourusername/ecoshop-capstone.git
git branch -M main
git push -u origin main
```

2. Create vercel.json

Create `vercel.json` in project root:

```
{
  "name": "ecoshop-capstone",
  "version": 2,
  "buildCommand": "exit 0",
  "devCommand": "npx http-server",
  "env": [],
  "public": true,
  "rewrites": [
    {
      "source": "/",
      "destination": "/capstone-index.html"
    }
  ],
  "routes": [
    {
      "src": "/(.*)",
      "dest": "/capstone-index.html"
    }
  ],
  "headers": [
    {
      "source": "/(.*)",
      "headers": [
        {
          "key": "Cache-Control",
          "value": "public, max-age=3600,
s-maxage=3600"
        }
      ]
    }
  ]
}
```

3. Deploy to Vercel

Option A: Via Web UI

1. Go to vercel.com
2. Click "Import Project"
3. Connect your GitHub account
4. Select `ecoshop-capstone` repository
5. Configure project:
 - Framework: `Other` (static)
 - Build Command: (leave blank)
 - Output Directory: `./`
6. Click "Deploy"
7. Wait for deployment to complete

Option B: Via CLI

```
# Install Vercel CLI
npm i -g vercel

# Login to Vercel
vercel login

# Deploy
vercel

# Follow prompts to configure and deploy
```

4. Verify Deployment

```
# Vercel will provide a URL like:
# https://ecoshop-capstone.vercel.app

# Test the URL in browser
# Verify all features work:
# - Login
# - Add products
# - Cart functionality
# - Orders
# - Responsive design
```

5. Configure Domain (Optional)

1. Go to Vercel dashboard
 2. Select your project
 3. Settings → Domains
 4. Add custom domain
 5. Follow DNS configuration steps
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Option 2: Netlify

Why Netlify?

- Easy drag-and-drop deployment
- GitHub integration
- Auto-publish on push
- Free tier with generous limits

- Built-in analytics

Step-by-Step Deployment

1. Prepare Repository

Same as Vercel option above.

2. Create `netlify.toml`

Create `netlify.toml` in project root:

```
[build]
  command = "echo 'Static site'"
  publish = "."

[context.production]
  publish = "."

[[redirects]]
  from = "/*"
  to = "/capstone-index.html"
  status = 200

[[headers]]
  for = "/*"
  [headers.values]
    Cache-Control = "public, max-age=3600, s-maxage=3600"
    X-Content-Type-Options = "nosniff"
    X-Frame-Options = "DENY"
```

3. Deploy to Netlify

Option A: Via Web UI (Drag & Drop)

1. Go to netlify.com
2. Click "Create a new site"
3. Drag your project folder onto the drop zone
4. Netlify automatically deploys
5. Access via provided URL

Option B: Via GitHub

1. Go to netlify.com
2. Click "Connect to Git"
3. Authorize GitHub
4. Select `ecoshop-capstone` repository
5. Configure:
 - Base directory: `./` (or root)
 - Build command: (leave blank for static)
 - Publish directory: `./`
6. Click "Deploy site"
7. Netlify watches for changes and auto-deploys

Option C: Via CLI

```
# Install Netlify CLI
npm install -g netlify-cli

# Login
```

```
netlify login

# Deploy
netlify deploy

# For production deployment
netlify deploy --prod
```

4. Verify Deployment

```
# Netlify provides URL like:
# https://ecoshop-capstone.netlify.app

# Test all features in browser
```

5. Configure Domain (Optional)

1. Dashboard → Site settings → Domain management
2. Add custom domain
3. Configure DNS records

Option 3: Cloudflare Pages

Why Cloudflare Pages?

- Ultra-fast global CDN
- Free tier with no limits

- Wrangler CLI for deployment
- Excellent DDoS protection
- Analytics included

Step-by-Step Deployment

1. Create wrangler.toml

```
name = "ecoshop-capstone"
type = "javascript"
main = "src/index.js"
compatibility_date = "2024-01-01"

[env.production]
name = "ecoshop-capstone-prod"

[[routes]]
pattern = "example.com/*"
zone_name = "example.com"
```

2. Prepare for Deployment

```
# Install Wrangler (Cloudflare CLI)
npm install -g wrangler

# Login to Cloudflare
wrangler login

# This opens browser for authentication
```

3. Deploy to Cloudflare Pages

Option A: Via GitHub Integration

1. Connect GitHub to Cloudflare
2. Select `ecoshop-capstone` repository
3. Configure build settings:
 - Framework preset: `None`
 - Build command: (leave blank)
 - Build output directory: `./`
4. Cloudflare Pages auto-deploys on push

Option B: Via Wrangler CLI

```
# Deploy directory
wrangler pages deploy .

# Wrangler creates Pages project
automatically
# Provides URL like: https://ecoshop-
capstone.pages.dev
```

4. Verify Deployment

Test all features in the provided URL.

5. Configure Domain

1. Cloudflare dashboard → Pages

2. Select project → Settings
 3. Custom domains
 4. Add your domain
-

Post-Deployment Verification

1. Functional Testing

- ☐ **Authentication**
 - ☐ Login with any credentials works
 - ☐ Logout works
 - ☐ User greeting displays
 - ☐ Session persists on page reload
- ☐ **Products**
 - ☐ All products display
 - ☐ Add product modal opens
 - ☐ Create new product works
 - ☐ Edit product works
 - ☐ Delete product works
- ☐ **Cart**
 - ☐ Add to cart updates badge
 - ☐ Cart view displays items
 - ☐ Update quantity works

- ☐ Remove item works
- ☐ Cart totals calculate correctly
- ☐ **Orders**
 - ☐ Checkout creates order
 - ☐ Order appears in history
 - ☐ Stock deducted from products
 - ☐ Cancel order works
 - ☐ Stock restored on cancel

2. Performance Testing

```
# Use Lighthouse (Chrome DevTools)
1. Open DevTools (F12)
2. Click Lighthouse tab
3. Generate report
4. Verify scores:
  - Performance: ≥ 90
  - Accessibility: ≥ 90
  - Best Practices: ≥ 90
  - SEO: ≥ 90
```

Or use online tools:

- [PageSpeed Insights](#)
- [WebPageTest](#)

3. Responsive Testing

Test on actual devices or emulation:

- ☐ Mobile (320px) - iPhone 5/SE
- ☐ Tablet (768px) - iPad
- ☐ Desktop (1024px+) - Full screen
- ☐ Extra wide (1440px+)

4. Browser Compatibility

Test on multiple browsers:

- ☐ Chrome (latest)
- ☐ Firefox (latest)
- ☐ Safari (latest)
- ☐ Edge (latest)

5. Security Verification

```
# Check security headers
curl -I https://your-deployment-url

# Should have headers:
# X-Content-Type-Options: nosniff
# X-Frame-Options: DENY
# Content-Security-Policy (if configured)
```

Environment Variables

If your app needs environment variables (future proofing):

Vercel

1. Project settings → Environment Variables
2. Add variable name and value
3. Select which environments
(Production/Preview/Development)
4. Redeploy

Netlify

1. Site settings → Build & Deploy → Environment
2. Add key/value pairs
3. Redeploy

Cloudflare Pages

1. Project settings → Environment variables
 2. Add variable name and value
 3. Redeploy
-

Monitoring & Analytics

Vercel Analytics

- Automatic with Vercel deployment
- Dashboard shows:
 - Page views
 - Unique visitors

- Geographic distribution

Netlify Analytics

1. Site settings → Analytics
2. Enable analytics
3. View insights in dashboard

Cloudflare Analytics

- Automatic with Cloudflare Pages
 - Dashboard shows:
 - Cache hits/misses
 - Bandwidth saved
 - Threat info
-

Troubleshooting

Issue: Blank Page After Deployment

Solution:

1. Check browser console (F12)
2. Look for 404 errors
3. Verify file paths are correct
4. Check if HTML file name matches deployment config

5. Ensure all JS files are in root or correctly referenced

Issue: localStorage Not Working

Solution:

1. Check if HTTPS is used (required in production)
2. Verify storage isn't disabled in browser
3. Check for QuotaExceededError in console
4. Clear browser cache and try again

Issue: Styles Not Loading

Solution:

1. Check if CSS is in index.html (inline)
2. Verify CSS has correct media queries
3. Clear browser cache
4. Hard refresh (Ctrl+Shift+R)

Issue: Forms Not Submitting

Solution:

1. Check console for JavaScript errors
2. Verify form IDs match JavaScript
3. Check event listener attachment
4. Verify preventDefault() is working

Issue: Slow Performance

Solution:

1. Check Lighthouse report
 2. Enable gzip compression (usually automatic)
 3. Optimize images if any
 4. Reduce CSS size (already minimal)
 5. Check for large localStorage data
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Auto-Deployment Configuration

Vercel (Automatic)

- Deploys on every push to main branch
- Preview deploys for pull requests
- Automatic rollback on errors

Netlify (Automatic)

- Deploys on every push to connected branch
- Preview deploys for pull requests
- Atomic deploys ensure consistency

Cloudflare (Manual or Automatic)

- Manual: `wrangler pages deploy .`
- Automatic: Via GitHub integration (if configured)

Custom Domain Setup

Step 1: Register Domain

- Use any registrar (GoDaddy, Namecheap, etc.)
- Note the nameservers

Step 2: Point Domain to Hosting

For Vercel:

1. Vercel dashboard → Project → Settings → Domains
2. Add custom domain
3. Follow DNS configuration (usually CNAME)

For Netlify:

1. Netlify dashboard → Settings → Domain management
2. Add custom domain
3. Update nameservers at registrar to Netlify's

For Cloudflare:

1. Use Cloudflare as DNS provider
2. Add domain to Cloudflare
3. Update nameservers at registrar

Step 3: Verify DNS

```
# Check DNS resolution
nslookup your-domain.com
dig your-domain.com

# Should resolve to your hosting provider
```

SSL/TLS Certificate

All modern hosting providers (Vercel, Netlify, Cloudflare) automatically provide:

- Free SSL/TLS certificates
- Auto-renewal
- HTTPS enforcement
- No manual configuration needed

Verify HTTPS:

```
# URL should show  padlock
https://your-deployment-url
```

Continuous Deployment Workflow

```
1. Clone repository locally
  git clone
  https://github.com/yourusername/ecoshop-
  capstone.git

2. Make changes
  git checkout -b feature/new-feature
  # Edit files

3. Commit and push
  git add .
  git commit -m "feat: Add new feature"
  git push origin feature/new-feature

4. Create pull request
  # GitHub shows deployment preview

5. Merge to main
  git merge feature/new-feature
  git push origin main

6. Auto-deployment triggers
  # Hosting provider deploys automatically
  # Your feature is now live
```

Rollback Procedures

Vercel

1. Dashboard → Deployments

2. Click on previous deployment
3. Click "Redeploy"
4. Confirms rollback

Netlify

1. Dashboard → Deploys
2. Click on previous deploy
3. Click "Restore"
4. Confirms rollback

Cloudflare

```
# List deployments
wrangler pages list deployments

# Redeploy specific version
wrangler pages rollback
```

Checklist for Live Deployment

Pre-Launch

- ☐ All features tested and working
- ☐ No console errors
- ☐ Lighthouse score > 90
- ☐ Mobile responsive verified

- ☐ Cross-browser tested
- ☐ Performance optimized

Deployment

- ☐ GitHub repository updated
- ☐ Deployment platform connected
- ☐ Environment configured
- ☐ Deployment completed successfully

Post-Launch

- ☐ Live URL verified
- ☐ All features tested on live
- ☐ Performance monitored
- ☐ Analytics enabled
- ☐ Team notified
- ☐ Submission completed

Support & Resources

- **Vercel Docs:** <https://vercel.com/docs>
- **Netlify Docs:** <https://docs.netlify.com>
- **Cloudflare Docs:**
<https://developers.cloudflare.com/pages/>
- **MDN Web Docs:** <https://developer.mozilla.org/>
- **Web.dev Best Practices:** <https://web.dev/>

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